

EPIC 4 – Administration, Finance & Royalty Management

Owner: Member 4

1. Epic Overview

Epic Name

Administration, Finance & Royalty Management

Owner

Member 4

Main Goal

Provide the publishing company with centralized financial management, royalty processing, administrative configuration, reporting, analytics, and executive oversight.

Epic 4 receives operational sales information from Epic 3 and combines it with author, contract, and publication information from the earlier epics to:

- Calculate royalties
- Generate royalty statements
- Manage royalty payments
- Manage financial transactions
- Generate financial reports
- Provide management analytics
- Maintain administrative master data
- Provide executive dashboards

Epic 4 represents the **financial and management layer** of the EPMS.

2. Business Purpose

Epic 4 replaces manual spreadsheets and separate financial records with a centralized financial and management system.

The module allows authorized staff to:

- Monitor revenue
 - Calculate author royalties
 - Generate royalty statements
 - Approve royalty payments
 - Record payments
 - Manage invoices
 - Record expenses
 - Generate financial reports
 - Analyze sales performance
 - Monitor royalty liabilities
 - Configure administrative settings
 - Manage category/genre master data
 - Provide executive-level analytics
-

3. Scope

Included

Administration

- Administrative dashboard
- Category management
- Genre management
- Publishing settings
- Royalty policy configuration
- Company settings
- Invoice settings
- System statistics
- Announcement management
- Read-only audit/activity monitoring

Finance

- Revenue monitoring
- Expense management
- Invoice management
- Payment records
- Financial reports
- Revenue reports
- Expense reports
- Profit analysis

- Sales reports
- Financial data export

Royalty Management

- Royalty agreements/terms
- Royalty rate configuration
- Royalty calculations
- Royalty statements
- Royalty payment approval
- Royalty payment records
- Royalty payment history
- Outstanding royalty balances

Analytics

- Revenue trends
- Sales trends
- Book performance
- Author performance
- Category performance
- Royalty analytics
- Inventory/sales statistics received from Epic 3
- Executive KPIs

4. Excluded

Epic 4 does **not** implement:

- Login
- Registration
- Authentication
- JWT
- Password management
- User management
- Role management
- RBAC infrastructure
- Author profile management
- Manuscript management
- Editorial review
- Book design
- Printing
- Warehouse operations
- Customer shopping cart

- Customer order processing
- Delivery management

These belong to the Shared Core, Epic 1, Epic 2, or Epic 3.

5. Important Domain Ownership

This is critical for preventing conflicts between members.

Domain	Owner
Users	Shared Core
Authentication	Shared Core
Roles/RBAC	Shared Core
Author profile	Epic 1
Manuscripts	Epic 1
Manuscript versions	Epic 1
Contracts	Epic 1
Editorial reviews	Epic 2
Books / production	Epic 2
Printing	Epic 3
Inventory	Epic 3
Customer orders	Epic 3
Deliveries	Epic 3
Sales transactions	Epic 3 operational source
Royalty calculation	Epic 4
Royalty payments	Epic 4
Financial reports	Epic 4

Domain	Owner
Expenses	Epic 4
Invoices	Epic 4
Financial analytics	Epic 4
Categories/Genres master data	Epic 4
Audit infrastructure	Shared Core

Important

Epic 4 may **view and report on** authors, books, customers, and sales.

It does **not** become the owner of those entities.

For example:

✗ "Epic 4 manages authors."

Correct:

☑ "Epic 4 retrieves author information from Epic 1 for financial and royalty purposes."

6. User Roles

Administrator

Can:

- Manage categories
- Manage genres
- Configure system settings
- Configure royalty policies
- Manage financial configuration
- Manage announcements
- View operational statistics
- View audit/activity information
- Access administrative reports

Cannot:

- Modify manuscripts
 - Modify editorial decisions
 - Modify inventory directly
 - Modify customer orders directly
-

Finance Officer

Can:

- Calculate royalties
 - Review royalty calculations
 - Approve royalty payments
 - Record royalty payments
 - Manage financial transactions
 - Manage expenses
 - Manage invoices
 - Generate financial reports
 - Export authorized financial reports
-

Executive

Can:

- View executive dashboard
- View revenue
- View sales analytics
- View royalty expenses
- View financial reports
- View business performance

Executive financial data is **read-only**.

Author

Can:

- View own royalty statements

- View own royalty calculations where permitted
- View own payment history
- Download own royalty statements

Cannot:

- View another author's financial information
- Modify royalty calculations
- Approve payments

7. Business Workflow

The complete financial workflow is:

```
Epic 3
Completed Sales
  ↓
Collect Sales Data
  ↓
Validate Completed Transactions
  ↓
Identify Applicable Royalty Agreement
  ↓
Calculate Royalty
  ↓
Generate Royalty Statement
  ↓
Finance Review
  ↓
Payment Approval
  ↓
Payment Scheduled
  ↓
Payment Processed
  ↓
Payment Completed
  ↓
Financial Reports
  ↓
Executive Analytics
```

8. Royalty Workflow

Royalty Agreement

DRAFT

↓
ACTIVE
↓
EXPIRED

An agreement may be superseded by a new agreement according to business rules.

It should not be physically deleted if historical calculations depend on it.

Royalty Calculation

SALES_RECEIVED
↓
CALCULATED
↓
REVIEWED
↓
APPROVED
↓
STATEMENT_GENERATED

Royalty Payment

PENDING
↓
APPROVED
↓
SCHEDULED
↓
PROCESSING
↓
PAID

An unsuccessful payment may enter:

FAILED

and be retried according to authorized business rules.

9. Financial Report Workflow

REQUESTED
↓
COLLECTING_DATA

↓
GENERATED
↓
REVIEWED
↓
FINALIZED

Once a financial report is finalized:

It becomes read-only.

10. Expense Workflow

RECORDED
↓
REVIEWED
↓
APPROVED
↓
POSTED

Rejected expenses should be explicitly recorded rather than deleted.

11. Permissions Matrix

Feature	Admin	Finance Officer	Executive	Author
View executive dashboard	☒	☒	☒	✗
Manage categories	☒	✗	✗	✗
Manage genres	☒	✗	✗	✗
Manage settings	☒	✗	✗	✗
Calculate royalties	✗/Authorized	☒	✗	✗
Review royalties	✗/Authorized	☒	Read-only	Own only
Approve royalty payment	✗	☒	✗	✗
Record royalty payment	✗	☒	✗	✗

Feature	Admin	Finance Officer	Executive	Author
View own royalty statement	✗	✗	✗	☒
Manage expenses	✗/Authorized	☒	Read-only	✗
Generate financial reports	☒	☒	Read-only	✗
Finalize financial reports	✗	☒	Read-only	✗
View financial analytics	☒	☒	☒	✗
View audit logs	☒	Authorized	Authorized/read-only	✗
Manage announcements	☒	✗	✗	✗

The exact role mapping must use the **Shared Core RBAC**.

12. Frontend Pages

Administration

- Admin Dashboard
- Category Management
- Genre Management
- System Settings
- Announcements
- Activity/Audit Viewer

Finance

- Finance Dashboard
- Revenue
- Expenses
- Invoices
- Financial Reports
- Payment History

Royalty

- Royalty Dashboard
- Royalty Agreements

- Royalty Calculations
- Royalty Statements
- Royalty Payments
- Payment History

Executive

- Executive Dashboard
- Business Analytics
- Revenue Analytics
- Sales Analytics
- Author Performance
- Book Performance

13. Executive Dashboard

The dashboard should display aggregated information such as:

- Total Revenue
- Monthly Revenue
- Completed Sales
- Books Sold
- Royalty Expenses
- Outstanding Royalties
- Paid Royalties
- Expenses
- Profit/Loss
- Best-Selling Books
- Top Performing Authors

Important

These are **dashboard features**, not a new dashboard infrastructure.

The page uses the Shared Core's layout and authentication/authorization mechanisms.

14. Royalty Dashboard

Displays:

- Pending calculations
 - Calculated royalties
 - Pending approvals
 - Scheduled payments
 - Paid royalties
 - Outstanding balances
 - Royalty expenses
 - Royalty trends
-

15. Royalty Agreement

Epic 4 needs the financial terms required to calculate royalties.

Possible information:

```
agreement_id
author_id
book_id
royalty_percentage
calculation_basis
effective_from
effective_to
status
created_at
updated_at
```

Important boundary

If the contractual agreement itself is owned by Epic 1, Epic 4 should **not create a second independent contract system**.

Instead:

```
Epic 1 Contract
      ↓
Financial/Royalty Terms
      ↓
Epic 4 Royalty Calculation
```

Epic 4 may maintain a `RoyaltyAgreement` financial representation/reference if the team has agreed that Epic 4 owns the financial terms.

The implementation must have **one authoritative source** for the contractual royalty terms.

16. Royalty Calculation

A basic royalty calculation can be represented as:

`Royalty Base × Royalty Percentage = Royalty Amount`

For example:

`Completed Sales Revenue = 100,000`

`Royalty Rate = 10%`

`Royalty = 100,000 × 10%`
`= 10,000`

The actual calculation must follow the agreed contract terms, including any deductions or exclusions.

17. Sales Period

Every royalty calculation must have a defined accounting period.

Example:

`2026-07-01 → 2026-07-31`

The system must prevent the same completed sale from being included in multiple finalized royalty calculations for the same agreement and accounting period.

This is essential to prevent **duplicate royalty payments**.

18. Database Ownership

Owned by Epic 4

- `royalty_agreements`
- `royalty_calculations`
- `royalty_payments`
- `payments`
- `invoices`
- `expenses`

- financial_reports
- categories
- genres
- system_settings
- announcements

Read-only / consumed

From Epic 1:

- Author
- Contract information
- Author identity/reference

From Epic 2:

- Book
- ISBN
- Publication metadata

From Epic 3:

- Completed sales
- Orders
- Order items
- Sales totals
- Inventory statistics
- Book sales counts

From Shared Core:

- Users
- Roles
- Audit log
- Notifications

19. Database Structure

RoyaltyAgreement

```
agreement_id
author_id FK/reference
book_id FK/reference
royalty_percentage
```

calculation_basis
effective_from
effective_to
status
created_at
updated_at

RoyaltyCalculation

calculation_id
agreement_id FK
sales_period_start
sales_period_end
books_sold
gross_sales
deductions
royalty_base
royalty_amount
status
calculated_by FK
calculated_at
created_at
updated_at

RoyaltyPayment

payment_id
calculation_id FK
author_id FK/reference
payment_amount
payment_method
payment_reference
payment_status
approved_by FK
approved_at
paid_at
created_at
updated_at

payment_reference must be unique.

20. Payment

payment_id
reference_number
payment_type
amount

payment_date
payment_method
status
description
created_at
updated_at

Possible payment types:

ROYALTY
EXPENSE
OTHER

The actual payment-processing integration can be implemented as appropriate for the university project.

21. Invoice

invoice_id
invoice_number
reference_type
reference_id
amount
tax_amount
total_amount
invoice_date
status
file_path
created_at
updated_at

Invoice numbers should be unique.

22. Expense

expense_id
category
description
amount
expense_date
status
approved_by FK
created_by FK
created_at
updated_at

23. FinancialReport

```
report_id
report_type
period_start
period_end
generated_by FK
generated_date
status
report_path
created_at
updated_at
```

24. Category

```
category_id
category_name
description
status
created_at
updated_at
```

Category names should be unique.

25. Genre

The original specification mentioned genres in the Master Prompt but omitted them from the database section.

They should therefore be explicitly defined:

```
genre_id
genre_name
description
status
created_at
updated_at
```

26. SystemSettings

```
setting_id
setting_key
setting_value
```

description
updated_by FK
updated_at

Examples:

currency
tax_rate
invoice_prefix
company_name
royalty_calculation_frequency

Important

Settings should not silently alter historical financial records.

A changed royalty rate affects **future calculations**, not already finalized calculations.

27. Announcements

If announcements are included in Epic 4:

announcement_id
title
content
status
published_at
expires_at
created_by FK
created_at
updated_at

This is an administrative content feature and must not replace the Shared Core's generic notification system.

28. Audit Log Boundary

Epic 4 does **not** own the audit log.

The Shared Core owns:

AuditLog

Epic 4 sends audit events to the shared audit service.

Examples:

- Royalty calculated
- Royalty approved
- Royalty payment approved
- Royalty payment recorded
- Expense created
- Expense approved
- Invoice generated
- Financial report generated
- Financial report finalized
- System setting changed
- Category created
- Category archived

Audit records must not be modified by Epic 4.

29. API Endpoints

Administration

GET	/api/admin/dashboard
GET	/api/categories
POST	/api/categories
PUT	/api/categories/{id}
PATCH	/api/categories/{id}/archive
GET	/api/genres
POST	/api/genres
PUT	/api/genres/{id}
PATCH	/api/genres/{id}/archive
GET	/api/settings
PUT	/api/settings
GET	/api/announcements
POST	/api/announcements
PUT	/api/announcements/{id}
PATCH	/api/announcements/{id}/archive

30. Royalty APIs

```
GET    /api/royalties
GET    /api/royalties/{id}

POST   /api/royalties/calculate
POST   /api/royalties/{id}/recalculate

GET    /api/royalties/{id}/statement
GET    /api/royalties/author/{authorId}
```

Only authorized finance personnel should access sensitive royalty calculations.

31. Royalty Payment APIs

```
GET    /api/royalty-payments
GET    /api/royalty-payments/{id}

POST   /api/royalty-payments/{id}/approve
POST   /api/royalty-payments/{id}/schedule
POST   /api/royalty-payments/{id}/process
POST   /api/royalty-payments/{id}/mark-paid
```

The endpoint design should ensure that a payment cannot be marked as paid without the required approval.

32. Finance APIs

```
GET    /api/finance/revenue
GET    /api/finance/expenses
POST   /api/finance/expenses
PUT    /api/finance/expenses/{id}

GET    /api/invoices
POST   /api/invoices/generate
GET    /api/invoices/{id}

GET    /api/payments
GET    /api/payments/{id}
```

33. Reporting APIs

```
GET    /api/reports
POST   /api/reports/generate
GET    /api/reports/{id}
GET    /api/reports/{id}/download
```

POST /api/reports/{id}/finalize

34. Analytics APIs

GET /api/analytics/dashboard
GET /api/analytics/revenue
GET /api/analytics/sales
GET /api/analytics/authors
GET /api/analytics/books
GET /api/analytics/royalties
GET /api/analytics/inventory

35. API Sequence – Royalty Calculation

The normal sequence is:

Epic 3
↓
Completed Sales
↓
Epic 4 retrieves sales
↓
Identify Author
↓
Identify Book
↓
Identify Active Royalty Agreement
↓
Validate Sales Period
↓
Calculate Royalty
↓
Store Calculation
↓
Generate Statement
↓
Finance Review
↓
Approve Payment
↓
Schedule Payment
↓
Process Payment
↓
Mark Paid

36. API Sequence – Financial Report

```
Request Report
  ↓
Validate User Permission
  ↓
Select Reporting Period
  ↓
Collect Sales Data
  ↓
Collect Expenses
  ↓
Collect Royalty Data
  ↓
Calculate Aggregates
  ↓
Generate Report
  ↓
Review
  ↓
Finalize
  ↓
Store
  ↓
Download
```

37. Frontend Responsibilities

Pages

- Admin Dashboard
 - Finance Dashboard
 - Royalty Dashboard
 - Royalty Agreements
 - Royalty Calculations
 - Royalty Statements
 - Royalty Payments
 - Expenses
 - Invoices
 - Financial Reports
 - Analytics Dashboard
 - Category Management
 - Genre Management
 - System Settings
 - Announcements
-

38. UI Components

- Revenue Cards
 - KPI Cards
 - Royalty Table
 - Payment Table
 - Expense Table
 - Invoice Table
 - Financial Summary
 - Revenue Charts
 - Sales Charts
 - Author Performance Chart
 - Book Performance Chart
 - Report Viewer
 - Export Button
 - Search
 - Filtering
 - Pagination
 - Date Range Selector
 - Status Badge
-

39. Business Rules

Royalty

1. Royalties are calculated only from completed sales.
2. Cancelled orders must not generate royalties.
3. Returned/refunded sales must be handled according to the royalty agreement.
4. The applicable royalty agreement must be active for the relevant sale period.
5. Each book/author relationship must use the correct royalty terms.
6. The same sale cannot be included twice in a finalized royalty calculation.
7. Finalized royalty calculations cannot be silently changed.
8. Recalculation requires authorization and an audit record.
9. Royalty payments require finance approval.
10. A royalty payment cannot be marked as paid without the required approval.
11. Every royalty payment requires a unique payment reference.

Financial

12. Completed sales are the basis for revenue reporting.
13. Financial reports use historical transaction data.
14. Finalized financial reports are read-only.
15. Financial transactions must be timestamped.
16. Invoice numbers must be unique.
17. Financial records must not be physically deleted if they are required for historical reporting.

18. Corrections should be handled through authorized adjustments/reversals rather than destructive edits.

Categories/Genres

19. Categories cannot be physically deleted while referenced by books or historical records.
20. Categories can be archived.
21. Archived categories cannot normally be assigned to new records.
22. Existing historical records retain their original category.

Settings

23. Changes to financial settings apply to future calculations.
24. Historical finalized calculations retain the values used at calculation time.
25. Unauthorized users cannot modify system settings.

40. Royalty Calculation Integrity

This is one of the most important controls in Epic 4.

Suppose July sales contain:

Book A
100 completed sales
Revenue = Rs. 100,000
Royalty = 10%

The royalty calculation is:

$100,000 \times 10\%$
= Rs. 10,000

Once this calculation is finalized, those same sales must not be included again in another finalized July calculation.

Therefore the system should maintain a relationship between:

Sales Transaction
↓
Royalty Calculation
↓
Royalty Payment

This prevents duplicate royalty payments.

41. Historical Data Rule

Historical financial information must remain reproducible.

For example, if the royalty rate changes from:

10% → 12%

the system must **not** recalculate last month's finalized royalty from 12%.

Historical calculations must preserve:

- Royalty rate used
- Sales period
- Sales base
- Deductions
- Royalty amount
- Calculation date
- Agreement reference

42. Business Exceptions

No active royalty agreement

→ Royalty calculation cannot be finalized.

Duplicate calculation

→ Reject calculation if the same sales period and agreement has already been finalized.

Cancelled sale

→ Exclude from completed-sales royalty base.

Returned sale

→ Apply the agreed return/refund treatment.

Payment without approval

→ Reject.

Duplicate payment reference

→ Reject.

Finalized report modification

→ Reject.

Archived category assigned to new book

→ Reject unless explicitly permitted.

Unauthorized financial export

→ Reject.

Author accessing another author's statement

→ Reject.

Missing sales data

→ Do not finalize the calculation until data integrity is resolved.

43. Security

Epic 4 consumes Shared Core authentication and RBAC.

It does not implement:

- Login
- Registration
- JWT
- Password encryption
- User management
- Role management
- RBAC infrastructure

Epic-specific security includes:

- Finance-role authorization

- Administrator authorization
 - Executive read-only access
 - Author ownership validation
 - Financial record protection
 - Royalty calculation authorization
 - Payment approval authorization
 - Report access control
 - Export authorization
 - IDOR prevention
 - Input validation
 - SQL injection protection
 - XSS protection
 - CSRF protection according to application architecture
 - Audit logging
 - Financial data confidentiality
-

44. Logging & Audit

The Shared Core Audit Service should record:

- Royalty agreement created
- Royalty agreement changed
- Royalty calculated
- Royalty recalculated
- Royalty approved
- Royalty statement generated
- Payment approved
- Payment scheduled
- Payment processed
- Payment marked paid
- Expense created
- Expense approved
- Invoice generated
- Report generated
- Report finalized
- Setting changed
- Category created
- Category archived
- Genre created
- Genre archived

Epic 4 does not create or maintain its own audit infrastructure.

45. Non-Functional Requirements

Accuracy

Financial calculations must use precise decimal arithmetic.

For Java/Spring Boot:

Use `BigDecimal` for monetary values.

Do **not** use floating-point `double` for financial calculations.

Transaction Integrity

Royalty calculation and payment state changes must be transactional.

Concurrency

Two finance officers must not be able to approve/process the same royalty payment simultaneously.

Auditability

Financial changes must be traceable.

Historical Integrity

Finalized financial data must remain immutable.

Performance

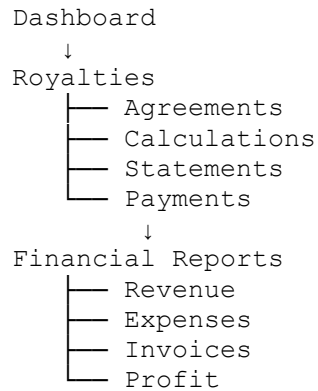
Dashboards should use optimized aggregate queries rather than loading all historical transactions into memory.

Security

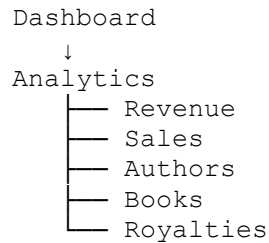
Financial information must only be exposed to authorized users.

46. UI Navigation

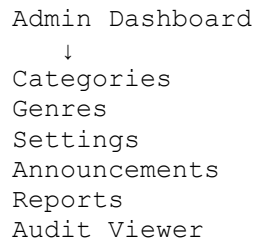
Finance Officer



Executive



Administrator



47. UI Components Reused From Shared Core

Epic 4 reuses:

- Navbar
- Sidebar
- User profile
- Shared dashboard layout

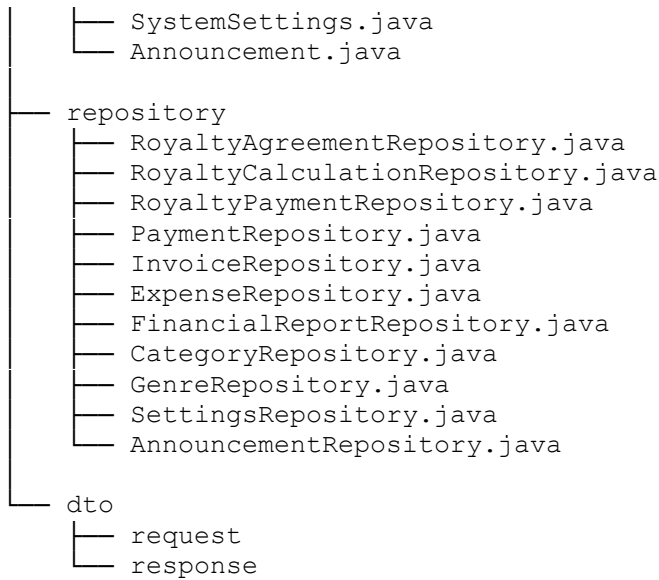
- Shared buttons
- Shared forms
- Shared tables
- Shared pagination
- Shared notification system
- Shared error handling
- Shared authorization mechanisms

Epic 4 must not create another dashboard framework.

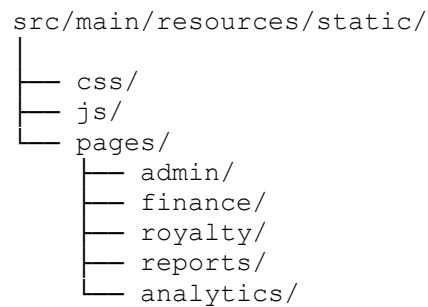
48. Folder Structure

Recommended backend:

```
com.epms
├── controller
│   ├── RoyaltyController.java
│   ├── RoyaltyPaymentController.java
│   ├── FinanceController.java
│   ├── ReportController.java
│   ├── AnalyticsController.java
│   ├── CategoryController.java
│   ├── GenreController.java
│   ├── SettingsController.java
│   └── AnnouncementController.java
├── service
│   ├── RoyaltyService.java
│   ├── RoyaltyPaymentService.java
│   ├── FinanceService.java
│   ├── ReportingService.java
│   ├── AnalyticsService.java
│   ├── CategoryService.java
│   ├── GenreService.java
│   ├── SettingsService.java
│   └── AnnouncementService.java
├── service/impl
└── entity
    ├── RoyaltyAgreement.java
    ├── RoyaltyCalculation.java
    ├── RoyaltyPayment.java
    ├── Payment.java
    ├── Invoice.java
    ├── Expense.java
    ├── FinancialReport.java
    ├── Category.java
    └── Genre.java
```



Frontend:



49. Dependencies

Before Member 4 begins implementation, the following must be available or have an agreed API contract.

Shared Core

- Authentication
- User
- Role
- RBAC
- Shared API response
- Shared security configuration
- Audit service
- Notification service

Epic 1

- Author information
- Contract information
- Royalty terms/reference

Epic 2

- Book information
- ISBN
- Publication information

Epic 3

- Completed sales
- Completed orders
- Order items
- Sales amounts
- Book sales counts
- Operational inventory statistics

50. Integration With Epic 1

Epic 4 receives:

- Author ID
- Author information
- Contract reference
- Royalty terms

Epic 4 uses these for royalty calculations and statements.

Important

Epic 4 does not edit the author's profile.

51. Integration With Epic 2

Epic 4 receives:

- Book ID
- Title
- ISBN
- Edition
- Publication metadata
- Author reference

This is used for:

- Sales reporting
- Book performance analytics
- Royalty calculations
- Financial reporting

Epic 4 does not edit editorial/production data.

52. Integration With Epic 3

Epic 4 receives:

- Completed sales
- Completed orders
- Order items
- Sales amount
- Book sales count
- Wholesale sales
- Inventory statistics
- Delivery completion information where required

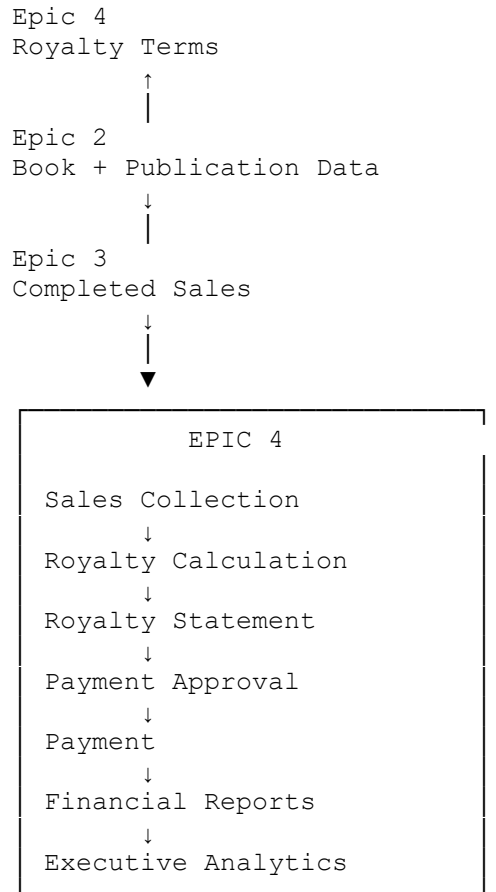
Important

Only completed/eligible transactions should contribute to revenue and royalty calculations according to the business rules.

53. Integration Data Flow

The overall flow becomes:

Epic 1
Author + Contract
↓



54. Acceptance Criteria

Epic 4 is complete when:

Administration

- ☐ Categories can be created.
- ☐ Categories can be edited.
- ☐ Categories can be archived.
- ☐ Genres can be managed.
- ☐ System settings can be managed by authorized administrators.
- ☐ Announcements can be managed.
- ☐ Shared audit events can be viewed where authorized.

Finance

- ☐ Revenue can be calculated from completed sales.
- ☐ Expenses can be recorded.
- ☐ Invoices can be generated.
- ☐ Payments can be recorded.
- ☐ Financial reports can be generated.
- ☐ Finalized reports are read-only.

Royalty

- ☐ Royalty agreements/terms can be accessed.
- ☐ Completed sales can be collected.
- ☐ Correct royalty rate is applied.
- ☐ Royalty calculations are stored.
- ☐ Duplicate calculations are prevented.
- ☐ Royalty statements can be generated.
- ☐ Payments require approval.
- ☐ Payment references are unique.
- ☐ Authors can view only their own statements.

Analytics

- ☐ Revenue dashboard works.
- ☐ Sales analytics works.
- ☐ Book performance works.
- ☐ Author performance works.
- ☐ Royalty analytics works.
- ☐ Executive dashboard works.

Integration

- ☐ Epic 1 author/contract information can be consumed.
 - ☐ Epic 2 book information can be consumed.
 - ☐ Epic 3 sales information can be consumed.
 - ☐ No domain ownership is duplicated.
 - ☐ Shared Core functionality is not recreated.
-

55. Testing Checklist

Royalty

- Calculate royalty
- Verify percentage
- Verify completed sales only
- Verify cancelled sales excluded
- Verify returned sales handling
- Verify correct agreement
- Prevent duplicate calculation
- Verify accounting period
- Verify recalculation authorization
- Generate statement

Payments

- Approve payment
- Reject unauthorized approval
- Prevent payment without calculation
- Prevent duplicate payment
- Validate payment reference
- Mark payment as paid
- Verify payment history

Finance

- Record expense
- Approve expense
- Generate revenue report
- Generate expense report
- Generate profit report
- Generate invoice
- Finalize report
- Prevent modification after finalization

Administration

- Create category
- Edit category
- Archive category
- Prevent deleting referenced category
- Create genre
- Update settings

- Verify settings affect future calculations only

Security

- Author cannot access another author's royalty statement
 - Executive cannot modify financial records
 - Unauthorized user cannot approve payments
 - Unauthorized user cannot change settings
 - Audit records cannot be modified
-

56. IE3121 Contribution

Business Architecture

Business Capabilities

- Financial Management
- Royalty Management
- Payment Management
- Financial Reporting
- Executive Management
- Business Analytics
- Administrative Configuration

Information Architecture

Entities:

- Royalty Agreement
- Royalty Calculation
- Royalty Payment
- Payment
- Invoice
- Expense
- Financial Report
- Category
- Genre
- System Settings
- Announcement

Application Architecture

- Finance Module

- Royalty Module
- Payment Module
- Reporting Module
- Analytics Module
- Administration Module

Technology Architecture

- Spring Boot
 - MySQL
 - REST APIs
 - Reporting Engine
 - PDF Export
 - Charting/Visualization
-

57. IE3151 Contribution

Epic 4 contributes:

- Epic Definition
 - User Stories
 - Product Backlog Items
 - Sprint Planning
 - Task Estimation
 - Acceptance Criteria
 - Test Cases
 - Sprint Reviews
 - Financial Documentation
 - Administrative Documentation
 - Technical Documentation
-

58. Deliverables

Member 4 is responsible for:

- Administration Module
- Royalty Management Module
- Royalty Calculation Engine
- Royalty Payment Module
- Finance Module
- Invoice Management

- Expense Management
- Financial Reporting Module
- Executive Dashboard
- Business Analytics Module
- Category/Genre Management
- System Settings
- REST APIs
- Database Tables
- Frontend Pages
- Unit Tests
- Integration Tests
- Technical Documentation

59. Verification Against Shared Core

Epic 4 does **not** implement:

- ✗ Login
- ✗ Registration
- ✗ Authentication
- ✗ JWT
- ✗ Password encryption
- ✗ User management
- ✗ Role management
- ✗ RBAC infrastructure
- ✗ Shared dashboard infrastructure
- ✗ Shared notification infrastructure
- ✗ Shared audit infrastructure

Epic 4 **consumes** those services.

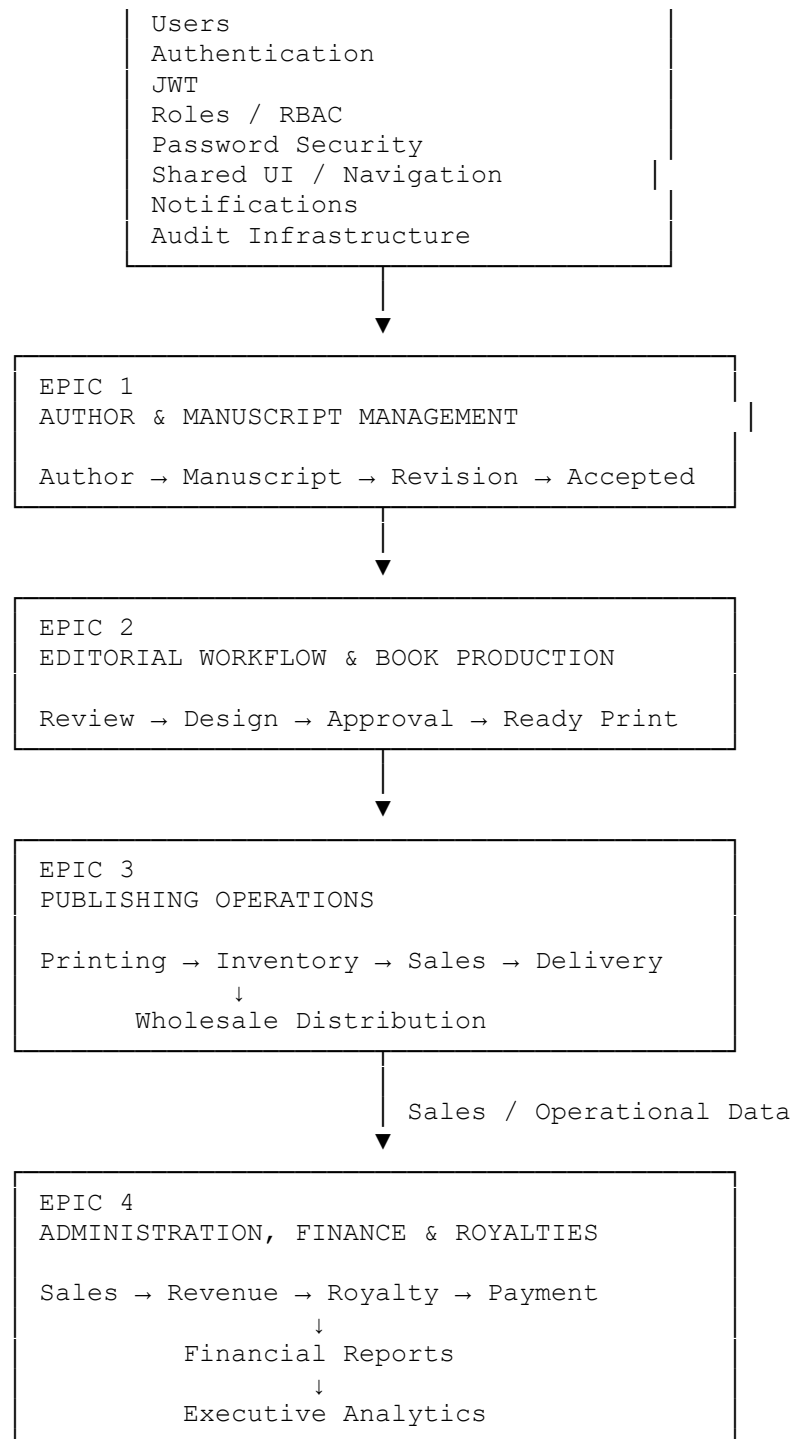
The only audit responsibility of Epic 4 is to **generate appropriate audit events and consume the shared audit service/viewer**.

60. Final Architecture Boundary

The final four-epic architecture should be:

SHARED CORE





61. Final Master Prompt — Epic 4

This is the **corrected version Member 4 should use**, rather than the original prompt:

I am developing Epic 4 of a university Enterprise Publishing Management System (EPMS).

Generate:

- Unit tests
- Integration tests
- API tests
- UI tests
- Royalty calculation tests
- Duplicate calculation tests
- Payment validation tests
- Financial validation tests
- Report generation tests
- Security tests
- Edge cases
- Negative test cases

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DOCUMENTATION

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Provide:

- Module Purpose
- Business Workflow
- Domain Ownership
- Database Design
- Entity Relationships
- REST API Documentation
- Royalty Calculation Logic
- Royalty Workflow
- Financial Workflow
- Backend Architecture
- Frontend Structure
- Folder Structure
- Validation Rules
- Exception Handling
- Security
- Audit Integration
- Business Constraints
- Reporting
- Analytics
- Integration Contracts
- Testing
- Future Enhancements

=====

IMPORTANT

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Do NOT redesign or recreate the Shared Core.

Do NOT create duplicate:

- Users
- Authors
- Books
- Orders

- Inventory
- Audit Logs
- Authentication
- Roles
- RBAC
- JWT
- Shared dashboard infrastructure

Respect the ownership boundaries between all four EPICS.

Epic 4 is responsible for:

Administration

Finance

Royalty Management

Payments

Financial Reporting

Analytics

Configuration

Act as a Senior Enterprise Software Architect and produce production-quality